

Write a C program for Find the Largest and Smallest Element in an Array

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int n;
4     if (scanf("%d", &n) != 1) {
5         return 0;
6     }
7     int arr[n];
8     for (int i = 0; i < n; i++) {
9         scanf("%d", &arr[i]);
10    }
11    int smallest = arr[0];
12    int largest = arr[0];
13    for (int i = 1; i < n; i++) {
14        if (arr[i] < smallest) {
15            smallest = arr[i];
16        }
17        if (arr[i] > largest) {
18            largest = arr[i];
19        }
20    }
21    printf("Smallest element: %d\n", smallest);
22    printf("Largest element: %d\n", largest);
23    return 0;
24 }
```

Input

```
5
10 25 3 48 17
```

Output

```
Smallest element: 3
Largest element: 48
```

Write a C program for Find the First and Last Occurrence of an Element

your Code

```
1 #include <stdio.h>
2 int main() {
3     int a[100], n, x, i, first = -1, last = -1;
4     scanf("%d", &n);
5     for (i = 0; i < n; i++)
6         scanf("%d", &a[i]);
7
8     scanf("%d", &x);
9     for (i = 0; i < n; i++) {
10         if (a[i] == x) {
11             if (first == -1)
12                 first = i;
13             last = i;
14         }
15     }
16     printf("First=%d\nLast=%d", first, last);
17     return 0;
18 }
```

Input

```
7
2 4 5 4 6 4 8
4
```

Output

```
First=1
Last=5
```

Write a program for binary search

ur Code

```
1 #include <stdio.h>
2 int binarySearch(int arr[], int n, int target) {
3     int low = 0;
4     int high = n - 1;
5     while (low <= high) {
6         int mid = low + (high - low) / 2;
7         if (arr[mid] == target) {
8             return mid;
9         }
10        if (arr[mid] < target) {
11            low = mid + 1;
12        } else {
13            high = mid - 1;
14        }
15    }
16    return -1;
17 }
18 int main() {
19     int n, target;
20     if (scanf("%d", &n) != 1) return 0;
21     int arr[n];
22     for (int i = 0; i < n; i++) {
```

Input

```
5
10 20 30 40 50
30
```

Output

```
2
```

Write a C program for palindrom

ur Code

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main() {
5     char str[100];
6     int i, len, is_palindrome = 1;
7
8     scanf("%s", str);
9     len = strlen(str);
10
11     for (i = 0; i < len / 2; i++) {
12         if (str[i] != str[len - 1 - i]) {
13             is_palindrome = 0;
14             break;
15         }
16     }
17
18     if (is_palindrome)
19         printf("Palindrome\n");
20     else
21         printf("Not Palindrome\n");
22 }
```

Input

stdin input

Output

